Embedding, zero-shot (n=199)

Masks + descriptions vs RGB + description (C5-C2) +1.5 [-2.5, +6.0]

Correct vs mismatched masks (C5-X1) +4.5 [+1.0, +8.5]

Masks vs RGB repeated (C5-X3) -2.5 [-7.0, +2.0]

Embedding, low-rank scorer, CV (n=799)

Masks + descriptions vs RGB + description (C5-C2) +2.6 [+0.4, +4.9]

Correct vs mismatched masks (C5-X1) +1.3 [-0.9, +3.5]

Masks vs RGB repeated (C5-X3) +2.1 [+0.4, +4.0]

Direct answering, Gemini 4 + GPT-5.6 + Qwen3-VL-32B

+ maps vs RGB +3.4 [+0.9, +5.9]

Correct vs mismatched maps -0.7 [-3.3, +1.9]

+ maps vs RGB repeated +2.8 [+0.4, +5.2]

+ descriptions vs RGB -1.3 [-5.1, +2.3]

Correct vs mismatched descriptions +3.4 [-0.7, +7.5]

Counterfactual: spatial facts reversed (text only)

GPT-5.6: reworded vs reversed +2.5 [-1.3, +6.3]

Gemini 4: reworded vs reversed +3.0 [-2.0, +8.0]

Qwen3-VL-32B: reworded vs reversed +5.5 [+0.5, +10.6]

Forced choice, "None" removed (GPT-5.6 + Qwen3-VL-32B)

+ maps vs RGB +0.5 [-2.5, +3.8]

Correct vs mismatched maps +0.0 [-3.0, +3.0]

Non-spatial text control: dataset scene description (GPT, Qwen-32B)

+ scene description vs RGB -8.0 [-13.8, -2.2]

Correct vs mismatched scene description +13.5 [+7.8, +19.1]

Prompting: reason about space first

GPT-5.6 (n=200): spatial analysis vs generic step-by-step -2.5 [-7.5, +2.5]

Qwen3-VL-32B (n=100): spatial analysis vs generic step-by-step -4.0 [-13.0, +5.0]

Published spatial fine-tunes vs their base (n=200)

SpatialLadder-3B vs Qwen2.5-VL-3B (paper: +23.5 spatial) -12.0 [-19.5, -4.5]

MindCube-RL-3B vs Qwen2.5-VL-3B (paper: +23.5 MindCube) -5.5 [-13.5, +2.0]

SpaceR-7B vs Qwen2.5-VL-7B (paper: +11.2 VSI-Bench) -0.5 [-5.5, +4.5]

Spatial-SSRL-7B vs Qwen2.5-VL-7B (paper: +3.9 avg) +1.5 [-3.0, +6.0]

Fine-tuning Qwen3-VL-8B (2 seeds, n=400)

Spatial LoRA vs non-spatial LoRA -6.5 [-12.0, -1.0]

Spatial LoRA x combined input, Qwen3-VL-8B (n=200)

Spatial LoRA + maps & descriptions vs base + RGB -9.8 [-16.5, -3.2]

Base: + maps & descriptions vs RGB -4.5 [-10.0, +1.0]

Spatial LoRA vs base, both given maps & descriptions -5.3 [-11.0, +0.5]

RL (GRPO) Qwen3-VL-8B, 90 steps

Norm reward vs untrained (reference) +8.3 [+4.0, +12.8]

Untrained: spatial-first prompt vs plain -17.0 [-24.0, -10.0]

RL (GRPO) Qwen3-VL-8B, matched 40 steps

Spatial + norm reward vs norm reward -7.0 [-13.5, -0.5]

Spatial format, norm reward vs norm reward -6.5 [-12.0, -0.5]

Spatial reward vs spatial format only -0.5 [-4.5, +3.5]

Pooled (random effects, midterm_meta.py)

All 11 spatial-specific contrasts -0.3 [-2.3, +1.7]

Effect on high- minus low-spatial-need items -1.4 [-4.6, +1.8]

−20 −10 0 10 20 30

Change in joint accuracy (percentage points, 95% CI)